EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY
DEPARTMENT OF SOFTWARE ENGINEERING

COURSE SYLLABUS

COURSE INFORMATION

Name	: **Introduction to Databases**
Course's Code	: CSE 301
Criteria for Registration	: Pre-requisite(s): CSE 201
Training Program	: Bachelor of Software Engineering
Program	: Software Engineering
Credits	: 4 (3,1)
Hours	: 60 hours (30 hours theory / 30 practice hours) + 75 hours self-study
Requirements	: Projectors, writing boards, practice computer rooms.
School/Department	: School of Computing and Information Technology – Department of Software Engineering.

COURSE DESCRIPTION

The course equips students with basic to specialized knowledge about databases, including the concept and architecture of database systems, detailed data modeling using Entity Relationship (ER) diagrams, relational models, conversion of ER models through relational models, etc the concepts of Keys and how keys are defined in database schemas, proficiency in the use of Structured Query Language (SQL), relational algebra, and calculations, along with database normalization to optimize performance and security.

COURSE OBJECTIVES

After completing the course, students must achieve the following objectives:

- Understand database concepts, applications, data models, schemas.

- Apply knowledge of combined entity models and relationship models to design Databases.

- Apply knowledge of relational algebraic language to answer queries.

CLO: Course Learning Outcome

- Applying the command structure of the SQL language, performing responses to data queries and installing types of integrity constraints on the data.

- Apply the knowledge of function dependencies, closing envelopes, and finding keys to standardize the database.

COURSE LEARNING OUTCOMES

		Implementation
CLO1	**Use** Entity Relationship model – ER and relationship model to design Database for the real problems.	PLO 1.2
CLO2	**Solve** queries in Relational Algebraic language, SQL, and how to express integrity constraints in formal language.	PLO 1.2
CLO3	**Apply** thinking to analyze problems and design ER models, relational models, and problem-solving in two languages: Relational Algebraic and SQL.	PLO 4.1 PLO 4.2
CLO4	**Evaluate** database schemas using data normalization forms.	PLO 5.1

MATERIALS

Books and Teaching Materials

[1]. Abraham Silberschatz, Henry Korth, and S. Sudarshan. 2020. *Database System Concepts (7th Edition)*. McGraw-Hill Publishers.

References

[2]. Raghu Ramakrishnan, Johannes Gehrke. 2003. *Database management systems (Third Edition)*. BPB publications.

[3]. Ivan Bayross. 2009. SQL, *PL/SQL the Programming Language of Oracle (4th Edition)*. McGraw-Hill Osborne Media.

[4]. Paul DuBois. 2013. *MySQL: Developer's Library (5th Edition)*. Addison-Wesley Professional .

COURSE OUTLINE

	Hours
Chapter 1. Introduction 1.1. Concept & Overview of Database Management System (DBMS) 1.2. Data Models, Database Languages, Database Administrator, Database Users 1.3. Three Schema architecture of DBMS.	(4 Lecture /0 Lab)
Chapter 2. Entity-Relationship Model 2.1. Introduction to Entity, Attribute, Types of Attributes 2.2. ER Model Notations, Strong and Weak Entity 2.3. Relationship Set, Mapping Cardinality, Participation Constraints	(4 Lecture /8 Lab)

CLO: Course Learning Outcome

2.4. Types of Keys
2.5. Extended E-R features: Generalization, Specialization, Aggregation

Chapter 3. Relational Model (6 Lecture / 0 Lab)
3.1. Structure of relational Databases
3.2. Relational Algebra
3.3. Operations in Relational Algebra
3.4. Extended Relational Algebra Operations

Chapter 4. SQL and Integrity Constraints (8 Lecture / 22 Lab)

4.1. Concept of Data Definition Language (DDL), Data Manipulation Language (DML), Data Control Language (DCL). Basic Structure

4.2. Set operations, Aggregate Functions, Null Values

4.3. Domain Constraints, Referential Integrity Constraints

4.4. Assertions, Views, Nested Subqueries
4.5. Stored Procedures and Triggers.

Chapter 5. Relational Database Design (8 Lecture / 0Lab)

5.1. Functional Dependency

5.2. Different anomalies in designing a Database

5.3. Normalization using functional dependencies

5.4. Decomposition

5.5. Normalization using multi-valued dependencies

SUM (30 Lecture /30Lab)

EVALUATION

Type		Content	Method	CLO	Weights
Regular	(1)	Practice 1 Practice 2 Practice 3 Practice 4 Practice 5 Practice 6 Practice 7	Practice on a computer	**CLO1** **CLO2** **CLO3**	**20%**
	(2)	ER model, querying data using relational algebraic language.	Midterm Exam (90-minute Practice Test)	**CLO1** **CLO2** **CLO3**	**30%**

CLO: Course Learning Outcome

Summary	(3)	ER Model, Relational Algebra, SQL Data Query, Function Dependency, Standard Form, etc.	Final exam (90-minute written test and computer practice)	**CLO1 CLO2 CLO3 CLO4**	**50%**
				SUM:	**100%**

SCHEDULE

Week 1

Hours: 4 Lecture / 0 Lab

Implement CLO1, CLO2, CLO3, CLO4.

Regular evaluation.

Content:

Chapter 1. Introduction

1.1. Concept & Overview of DBMS
1.2. Data Models, Database Languages, Database Administrator, Database Users
1.3. Three Schema architecture of DBMS.

Evaluation Activities:

1. Listen to the lecture and discuss the content of chapter 1.

Week 2

Hours: 4 Lecture / 4 Lab

Implement CLO1, CLO3.

Regular evaluation (1).

Content:

Chương 2. Entity-Relationship Model

2.1. Introduction to Entity, Attribute, Types of Attributes
2.2. ER Model Notations, Strong and Weak Entity
2.3. Relationship Set, Mapping Cardinality, Participation Constraints
2.4. Types of Keys
2.5. Extended E-R features: Generalization, Specialization, Aggregation

Evaluation Activities:

1. Listen to lectures, discuss, do class assignments.
2. Practice 1.

Week 3

Hours: 4 Lecture / 4 Lab

Implement CLO2, CLO3.

Regular evaluation (1).

Content:

Chương 3. Relational Model

3.1. Structure of relational Databases
3.2. Relational Algebra
3.3. Operations in Relational Algebra

Evaluation Activities:

CLO: Course Learning Outcome

1. Listen to lectures, discuss, do class assignments.
2. Practice 2.

Week 4

Hours: 4 Lecture / 4 Lab

Implement CLO2, CLO3

Regular evaluation (1)

Content:

Chương 3. Relational Model

3.4. Extended Relational Algebra Operations

Chương 4. SQL and Integrity Constraints

4.1. Concept of DDL, DML, DCL. Basic Structure

Evaluation Activities:

1. Listen to lectures, discuss, do class assignments.
2. Practice 3

Week 5

Hours: 4 Lecture / 4 Lab

Implement CLO2, CLO3

Regular evaluation (1)(2)

Content:

Chương 4. SQL and Integrity Constraints

4.2. Set operations, Aggregate Functions, Null Values

4.3. Domain Constraints, Referential Integrity Constraints

4.4. Assertions, views, Nested Subqueries

Evaluation Activities:

1. Listen to lectures, discuss, do class assignments.
2. Practice 4
3. Midterm exams

Week 6

Hours: 4 Lecture / 4 Lab

Implement CLO2, CLO3, CLO4

Regular evaluation (1)(2).

Content:

Chương 4. SQL and Integrity Constraints

4.5. Stored procedures and triggers

Chương 5. Relational Database Design

5.1. Functional Dependency

5.2. Different anomalies in designing a Database

5.3. Normalization using functional dependencies

5.4. Decomposition

5.5. Normalization using multi-valued depedencies

Evaluation Activities:

1. Listen to lectures, discuss, do class assignments.
2. Practice 5

CLO: Course Learning Outcome

Week 7

Hours: 4 Lecture / 4 Lab

Implement CLO2, CLO3, CLO4.

Regular evaluation (1)

Content:

Chương 5. Relational Database Design

5.2. Different anomalies in designing a Database

5.3. Normalization using functional dependencies

5.4. Decomposition

Evaluation Activities:

1. Listen to lectures, discuss, do class assignments.
2. Practice 6

Week 8

Hours: 2 Lecture / 4 Lab

Implement CLO4

Regular evaluation (1)

Content:

Chương 5. Relational Database Design

5.5. Normalization using multi-valued dependencies

Evaluation Activities:

1. Group assignments, discussions, classwork
2. Practice 7

Week 9

Hours: 0 Lecture / 2 Lab

Implement CLO1,CLO2, CLO3, CLO4

Regular evaluation (1)

Content: Course review

1. Review key content, answer questions

Evaluation Activities:

1. Exchange, discuss, and answer questions related to the COURSE.

COURSE POLICIES

- Absence of more than 20% of sessions: students are banned from taking the exam.
- Students who submit assignments late in time: deduct 30% of the points.

INSTRUCTORS

Lecturer 1

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Academic Title: Master

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Lecturer 2

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CLO: Course Learning Outcome

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Binh Duong, the day ... month... 20..

Dean (*Sign and write full name*)	**Head of Department** (*Sign and write full name*)	**Lecturer 1** (*Sign and write full name*)	**Lecturer 2** (*Sign and write full name*)
Narayan DEBNATH C	**Narayan DEBNATH C**	**Vương Quốc Dương**	

CLO: Course Learning Outcome

ADDENDUM

APPENDIX I. SCORING RUBRIC

1. RUBRIC ASSIGNMENTS

This rubric is used to perform assessment number (1)

CRITERIA	CLO	10% points	50% points	80% points	100% points	MAXIMUM POINTS
1. Deduce, find solutions to solve the requirements of the exercise	CLO1,CLO2, CLO3	There is analysis and reasoning but no exercises have been solved	Do 50% of the exercises	I can do the exercises but the results are not accurate	Do all the exercises and get the correct results	**100**
					Sum:	**100**

2. RUBRIC MID-TERM EXAM

Use to perform digital activities (2)

Criteria	CLO	Know	Understand	Apply	Analyze	Evaluate	Creative	Total questions	Point
Apply ER model knowledge to design database according to problem requirements	CLO1, CLO3			*1*				*1*	*60*
Apply relational algebraic language to execute query requests.	CLO2, CLO3			*1*				*1*	*40*
							Sum	*2*	**100**

GRADE SCALE

CLO: Course Learning Outcome

Question 1 (60 points): Apply ER model knowledge to design database according to problem requirements. **60**

Question 2 (40 points): Apply relational algebraic language to perform query requests. **40**

Sum 100

3. RUBRIC FINAL EXAM

Use to perform digital activities (3)

Criteria	CLO	Know	Understand	Apply	Analyze	Evaluate	Creative	Total questions	Point
Apply ER model knowledge to design database according to problem requirements	CLO1, CLO3			*1*				*1*	*30*
Apply the dependency function to find the envelope, the minimum cover, the standard form of the relationship.	CLO4			*1*				*1*	*10*
Apply relational algebraic language to execute query requests.	CLO2, CLO3			*1*				*1*	*20*
Apply the SQL language to execute query requests.	CLO2, CLO3			*1*				*1*	*40*
							Sum	*4*	**100**

GRADE SCALE

Question 1 (10 points): Apply ER model knowledge to design database according to problem requirements. **30**

Question 2 (20 points): Apply the dependent function to find the envelope, the minimum cover, the standard form of the relationship. **10**

Question 3 (30 points): Apply relational algebraic language to perform query requests. **20**

CLO: Course Learning Outcome

Question 4 (40 points): Apply the SQL language to execute query requests. **40**

Sum 100

CLO: Course Learning Outcome